E-COMMERCE CHECKOUT FUNNEL OPTIMIZATION

BRD DOCUMENT

2026

LATESH ANJARLEKAR
Business Analyst

1. Project Overview

This project focuses on analyzing and optimizing the e-commerce checkout funnel to improve user conversion and reduce drop-offs. The analysis is based on user-level behavioural event data across key funnel stages, with the goal of identifying friction points and implementing targeted improvements to increase completed transactions and revenue.

---

2. Business Problem

A significant proportion of users entering the checkout funnel are dropping off before completing a purchase, leading to revenue leakage.

- Estimated current funnel conversion rate: **1.2–1.8%**
- Industry benchmark: **2–3%**
- Drop-offs are concentrated in:
    - Cart → Checkout transition
    - Checkout → Purchase completion

Impact:

- High cart abandonment
- Lost revenue from high-intent users
- Inefficient conversion of acquired traffic

---

3. Objectives

- Increase purchase conversion rate by **15–25%**
- Reduce checkout-stage drop-offs by **20%**
- Improve user progression across funnel stages
- Identify behavioural and device-level friction patterns
- Enable data-driven product improvements

---

4. Scope

In Scope:

- Checkout funnel analysis (view → cart → checkout → purchase)
- User behaviour tracking across funnel stages
- Device and country segmentation
- Conversion and drop-off analysis
- Checkout experience optimization

Out of Scope

- Marketing acquisition channels

- Pricing strategy changes
- Inventory management
- Logistics and delivery optimization

---

5. KPIs

North Star Metric:

- **Purchase Conversion Rate = Purchases / Product Views**

Supporting Metrics:

- Add-to-Cart Rate
- Checkout Initiation Rate
- Checkout Completion Rate
- Drop-off % at each stage
- Revenue per User
- Average Order Value (AOV)
- Device-wise Conversion Rate

---

6. Dataset Explanation

Field Name	Description	Business Purpose
user_pseudo_id	Unique user identifier	Track user journey
event_name	Funnel action	Define funnel stage
event_timestamp	Event time	Sequence events
device	Device type	Analyze device behaviour
country	User location	Regional insights
transaction_id	Purchase identifier	Validate conversions
purchase_revenue	Revenue value	Revenue analysis

---

7. Stakeholder Register

Stakeholder	Role	Responsibility
Product Manager	Decision Maker	Defines product direction
Business Analyst	Analyst	Data analysis & insights
Data Analyst	Support	SQL & data extraction
UX Designer	Contributor	Improve checkout UI
Engineering Team	Execution	Implement changes

---

8. High-Level Business Process (AS-IS)

- User lands on product page

- Views product details
- Adds item to cart
- Proceeds to checkout
- Enters shipping & payment details
- Completes purchase

Drop-offs can occur at:

- Cart stage (intent loss)
- Checkout stage (friction)
- Payment stage (failure or hesitation)

---

9. Business Requirements

BR-1: Funnel Visibility

The business requires visibility into user progression across funnel stages in order to identify drop-off points.

BR-2: Drop-off Identification

The business requires identification of high drop-off stages in order to prioritize optimization efforts.

BR-3: Segmentation Analysis

The business requires segmentation by device and geography in order to detect behavioural differences.

BR-4: Checkout Friction Analysis

The business requires analysis of checkout-stage behaviour in order to identify usability or process-related issues.

BR-5: Revenue Impact Measurement

The business requires mapping of funnel performance to revenue in order to quantify business impact.

---

10. Assumptions

- Each user ID represents a unique user
- Events are tracked accurately
- Purchase events indicate successful transactions
- Funnel steps are sequential
- No major tracking gaps exist

---

11. **Constraints**

- Limited visibility into user intent beyond events
- No session-level granularity (if applicable)
- No external behavioural data (ads, campaigns)

---

12. **Dependencies**

- Data availability and accuracy
- Event tracking implementation
- Stakeholder alignment on KPIs

---

13. **Risks**

- Missing or inconsistent event tracking
- Duplicate user records affecting accuracy
- Time zone inconsistencies in timestamps
- Incorrect revenue attribution

---

14. **Sign-Off**

Role	Name	Signature	Date
Product Manager			
Business Analyst			
Engineering Lead			